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REGULAR PAPER

Vine copula classifiers for the mind reading problem

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Abstract In this paper we introduce vine copulas to model probabilistic dependencies in supervised classification problems. Vine copulas allow the representation of the dependence structure of multidimensional distributions as a factorization of bivariate pair-copulas. The flexibility of this model lies in the fact that we can mix different types of pair-copulas in a factorization, which allows covering a wide range of types of dependencies, i.e., from independence to much more complex forms of bivariate correlations. This property motivates us to use vine copulas as classifiers, particularly for problems for which the type and strength of bivariate interactions between the variables show a great variability. This is the case of brain signal classification problems where information is represented as multiple time series, each one recorded from different brain region. Our experimental results on a real-word Mind Reading Problem reveal that vine copula-based classifiers perform competitively compared to the four best classification methods presented at the Mind Reading Challenge Competition 2011.

Keywords Vine · Copulas · Vine copulas classifiers · Mind reading

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1 Introduction

Copulas are increasingly popular in machine learning due to the fact that they provide a flexible tool to build multivariate distributions from marginals which are modeled by different distributions, and a function (the copula) that links these marginals. This coupling feature allows the modeling of marginals and the dependence model (from different copula families) separately.

Despite the generality of the copula-based framework, building high-dimensional copulas is a difficult problem [1]. While there is a variety of bivariate copula families that cover a wide range of different types of correlations, the number of standard higher dimensional joint copulas (such as the multivariate Gaussian, Student *t* and Archimedian copulas) is quite restricted. In addition, these copulas express the same type of dependence between all pairs of variables, which may be too restrictive if the variables interact differently (e.g. linear and non-linear).

To overcome the above mentioned limitations, in [16] a method is developed that allows us to approximate multivariate distributions using simple building blocks called pair-copulas. To organize these constructions, a graphical model called regular vine (Rvine) that allows decomposing a multivariate joint copula into a cascade of bivariate copulas (pair-copulas) is introduced in [2,3]. Vine copulas offer a more flexible way to represent multivariate dependence structures than the standard multivariate copulas, since pair-copulas belonging to different families can be mixed in a Rvine. In this paper, we focus on a particular type of Rvine called drawable vine (Dvine).

Previous works on the use of copula functions as classifiers have been limited to a few papers. In [8], a graphical model called copula bayesian network (CBN) is introduced and applied to several supervised classification benchmark problems. CBN integrates the copula and Bayesian network (BN) frameworks. Furthermore, it captures the multivariate dependence structure through a set of local copula densities associated with the nodes of the BN. The BIC score is used to select between the Gaussian and Clayton copulas to model the local copulas. In [32,33], a Bayes classifier based on trivariate Gaussian copulas is used for supervised color pixel classification to decide if the pixel belongs to either the foreground or the background class.

The properties of vine copulas motivate us to incorporate them into a supervised classification scheme. To the best of our knowledge, these models have not been studied in this context. Our main goal is then to investigate the suitability of vine copulas for solving supervised classification tasks, particularly for problems for which the type and strength of bivariate interactions between variables show a great variability. This is the case of brain signal classification problems. The potential existence of complex, diverse, and intricate patterns of interactions between the variables of the mind decoding or mind reading problem (MRP) made us to select vine copulas as a promising candidate approach to the problem.

MRP consists of decoding the original stimuli (e.g., seen, heard) received by a subject from the analysis of his/her brain signals. The type of stimuli as well as the methods used to register the subject's brain activity may differ (e.g., electroencephalography (EEG) vs. magnetoencephalography (MEG) [23]).

In this paper we focus on a particular experimental benchmark for MRP. The benchmark was supplied by the MEG Mind Reading Challenge Competition [20] to be used in the benchmarking of brain decoding methods. Here, the MEG activity of a subject who was watching videos of different types (or content) was recorded. The classification problem consists of determining which type of video (among the five possible) the subject has watched by analyzing the brain signals. From now on, when we refer to MRP we consider the particular problem treated in this paper.

Machine learning algorithms have been used to decode the information contained in the brain signals and recognize a stimulus received by the subject or an intended action (e.g., decide between right and left movements) [22]. From the point of view of machine learning, MRP can be described as a multiclass classification problem with five classes. However, the problem is considerably difficult due to the high noise of the brain data, which corresponds to a set of multivariate time series with a large number of variables. Moreover, there is also a variability between the distribution of the training and test data, a problem usually called *covariate shift* [36].

At the Mind Reading Competition 2011, eight different classification methods were evaluated for the brain decoding problem using MEG data. These methods included regularized logistic regression [15], ensembles of classifiers [34], multiclass Gaussian process [19], and others. In this paper we follow a complete different approach based on the use of regular vines, a class of probabilistic graphical models based on copulas [24].

We introduce the vine copula classifier in the framework of probabilistic classifiers, where the dependence between the variables and the class is expressed as a vine copula. In our proposal, for each class, a vine copula is used to estimate the probability distribution of the training samples that belong to the class. Then, the learned models are used to predict the probability of the unlabeled instances, and for each instance the class whose corresponding model has assigned the highest probability to the instance is selected.

Another contribution of our work has to do with the dimensions of the problem treated here. To the best of our knowledge, vine copulas have been usually restricted to no more than 20 variables. MRP has 408 MEG brain signal features, which gives us the opportunity to test the feasibility of vine copulas to work with real larger problems.

The remainder of this paper is organized as follows: Sect. 2 offers an introduction to copulas and discusses bivariate copula families. Section 3 presents both the pair-copula construction method and the Dvine copula model to approximate higher dimensional distributions. In Sect. 4 we develop the vine copula-based classification approach put forward in this paper. In Sect. 5 we describe MRP. Sections 6 and 7 present the experimental framework and discuss the obtained results. Section 8 concludes with a summary and an outlook to possible future research.

2 Introduction to copulas

Let $\mathbf{X} = (X_1, \ldots, X_n)$ be an n-dimensional random vector with joint density function $f : \mathbb{R}^n \to [0, \infty)$ and cumulative distribution function (cdf) $F : \mathbb{R}^n \to [0, 1]$. Further, let $F_i : \mathbb{R} \to [0, 1]$, $i = 1, \ldots, n$, be the corresponding marginal distributions of X_i, $i = 1, \ldots, n$.

Definition 1 (*Copula*) A copula $C : [0, 1]^n \to [0, 1]$ is an n-dimensional distribution function on $[0, 1]^n$ with uniformly distributed margins.

From the above definition, if F_i, $i = 1, \ldots, n$, are univariate distribution functions, the copula $C\left(F_1(x_1), \ldots, F_n(x_n)\right)$ is a multivariate distribution function with marginals F_i ($U_i = F_i(X_i)$ is a uniform random variable).

A key result of copula theory, Sklar's theorem [37], states how a copula function is related to a joint distribution function.

Theorem 1 (Sklar's theorem) *Let F be an n-dimensional distribution function with continuous marginals $F_1, \ldots, F_n$. Then it has a unique copula representation*

$$F(x_1, \ldots, x_n) = C(F_1(x_1), \ldots, F_n(x_n)), \\ \times (x_1, \ldots, x_n) \in \mathbb{R}^n. \quad (1)$$

Moreover, the following corollary is obtained from (1):

Corollary 1 *Let F be an n-dimensional distribution function with continuous marginals $F_1, \ldots, F_n$ and copula C satisfying (1). Then, for any $\mathbf{u} = (u_1, \ldots, u_n)$ in $[0, 1]^n$ we have*

$$C(u_1, \ldots, u_n) = F\left(F_1^{-1}(u_1), \ldots, F_n^{-1}(u_n)\right), \quad (2)$$

where F_i^{-1} is the inverse of F_i.

For an absolutely continuous F with strictly increasing, continuous marginal densities $F_1, \ldots, F_n$, the joint density function f is obtained from (1) as

$$f(x_1, \ldots, x_n) = c(F_1(x_1), \ldots, F_n(x_n)) \cdot \prod_{i=1}^{n} f_i(x_i), \quad (3)$$

where c and f_i denote the corresponding density functions of C and F_i, respectively. From (3) we can clearly see that copulas allow us to separately model the marginals and the joint dependence structure.

2.1 Examples of bivariate copulas

Pair-copulas are building blocks for constructing multivariate vine copula models. In this section we present four of the most common bivariate copulas. They express several types of dependencies that may be perceived in the variation of some features such as tail dependence, symmetry, and type of the correlation, among others [24].

2.1.1 Bivariate product copula

Two random variables X_1 and X_2 with continuous distribution functions F_1 and F_2 and joint distribution function F are independent if and only if $F(x_1, x_2) = F_1(x_1).F_2(x_2)$ for all $x_1, x_2 \in \mathbb{R}$. The structure of such a relationship is given by the independence or product copula, whose distribution and density functions are defined as

$$C(u, v) = u.v \quad (4)$$

$$c(u, v) = 1 \quad (5)$$

Scatter plot of bivariate independent data is shown in Fig. 1.

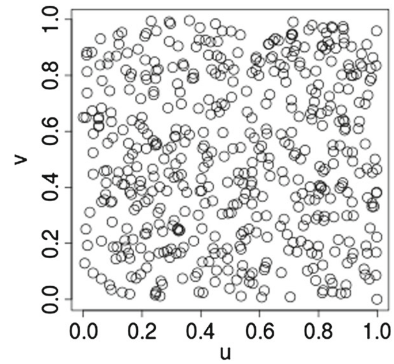

Fig. 1 Scatter plot resulting from simulating the product copula

2.1.2 Bivariate Gaussian copula

From (2), the bivariate Gaussian copula has distribution and density functions

$$C(u, v) = \Phi_\rho\left(\Phi^{-1}(u), \Phi^{-1}(v)\right) \quad (6)$$

$$c(u, v; \rho) = \frac{1}{\sqrt{1-\rho^2}} \exp\left(-\frac{\rho^2\left(x^2+y^2\right) - 2\rho x y}{2\left(1-\rho^2\right)}\right) \quad (7)$$

respectively, where Φ_ρ denotes the bivariate normal distribution function ($\mu_u = \mu_v = \sigma_u = \sigma_v = 0$) with the correlation coefficient $\rho \in (-1, 1)$, and Φ^{-1} is the inverse cumulative distribution function of a standard normal Φ.

2.1.3 Bivariate Clayton copula

The bivariate Clayton copula has distribution and density functions

$$C(u, v) = \left(u^{-\delta} + v^{-\delta} - 1\right)^{-1/\delta} \quad (8)$$

$$c(u, v) = (1-\delta)(uv)^{-1-\delta}\left(u^{-\delta} + v^{-\delta} - 1\right)^{-\frac{1}{\delta}-2}, \quad (9)$$

respectively, with dependence parameter $\delta \in (0, \infty)$.

2.1.4 Bivariate Gumbel copula

The bivariate Gumbel copula has distribution and density functions

$$C(u, v) = \exp\left(-\left((-\log u)^\delta + (-\log v)^\delta\right)^{\frac{1}{\delta}}\right) \quad (10)$$

$$c(u, v) = \frac{1}{C(u, v)}\left(-\left((-\log u)^\delta + (-\log v)^\delta\right)^{\frac{2}{\delta}-2}\right) \\ \cdot (\log u \cdot \log v)^{\delta-1} \\ \cdot \left(1 + (\delta-1)\left((-\log u)^\delta + (-\log v)^\delta\right)^{\frac{1}{\delta}}\right) \quad (11)$$

with dependence parameter $\delta \in [1, \infty)$.

Figure 2 shows Gaussian, Clayton and Gumbel bivariate samples for different values of the dependence parameter.

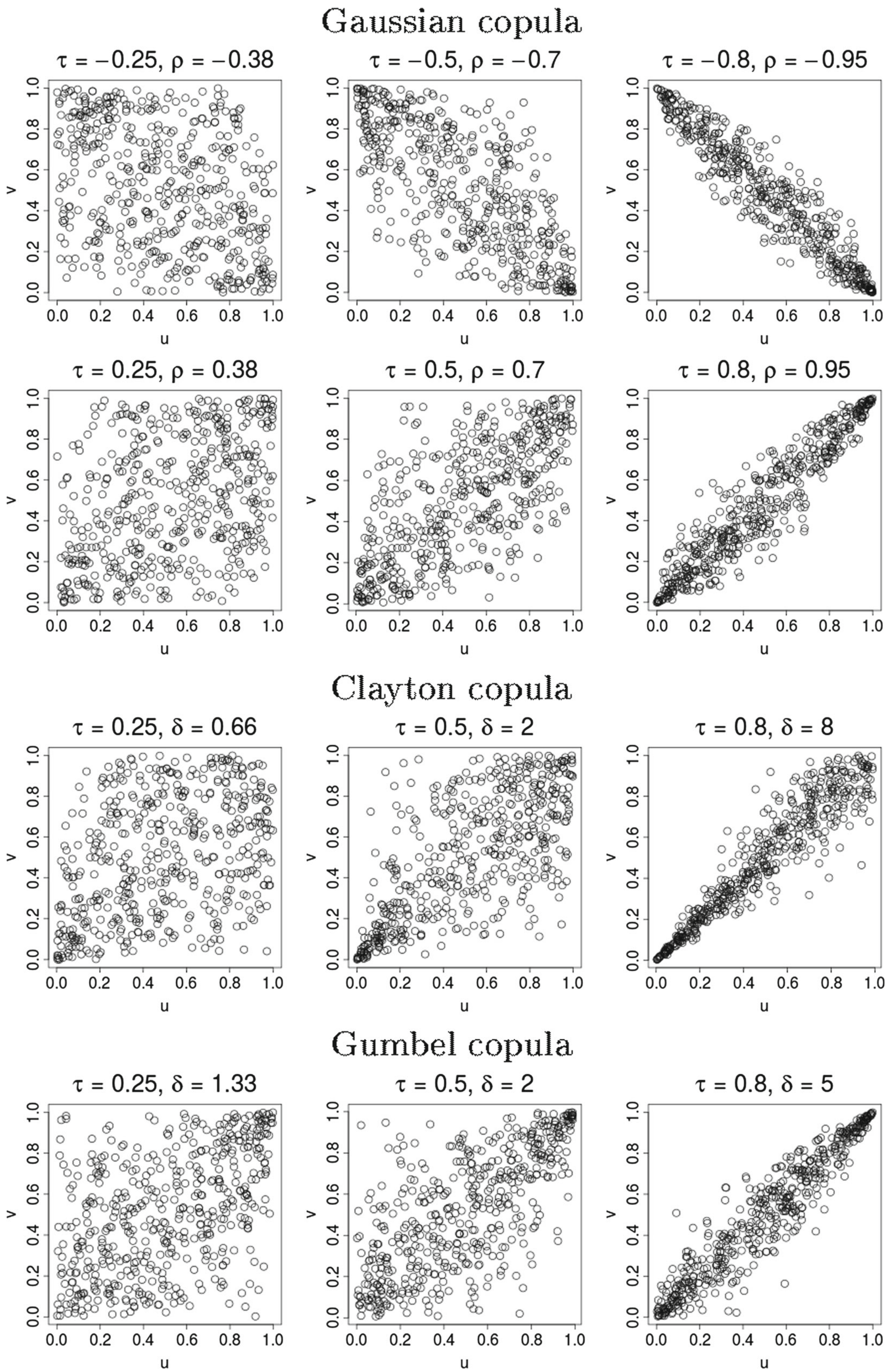

Fig. 2 Scatter plots of 500 points sampled from different types of bivariate copulas. The bivariate Gaussian copula is symmetric and neither lower nor upper tail dependent. It can represent positive/negative correlations. The bivariate Clayton copula is lower but not upper tail dependent. The bivariate Gumbel copula is upper tail dependent, but not lower. These two copulas are both asymmetric and can only describe positive correlations

3 Vine copulas

The type of the relations between n random variables cannot be modeled with single standard multivariate copulas (e.g. the Gaussian, Student t and Archimedian copulas). These copulas express the same type of dependence between all pairs of variables, which is too restrictive when the variables are correlated in different ways.

Pair-copula constructions (PCCs) solve this matter using two-dimensional copulas. The modeling scheme is based on a decomposition of a multivariate density into a cascade of (conditional and unconditional) bivariate copulas, called pair-copulas. PCC models provide a more flexible way of pair-wise dependence modeling than multivariate copulas, since pair-copulas belonging to different families can be mixed in the same decomposition.

In this section we define such a construction following the presentation in [1,6].

Consider n random variables $\mathbf{X} = (X_1, \ldots, X_n)$ with a joint density function $f(x_1, \ldots, x_n)$. This density can be factorised using the chain rule as

$$f(x_1, \ldots, x_n) = f(x_1) . f(x_2 \mid x_1) \cdot f(x_3 \mid x_1, x_2) \cdot \cdots \cdot f(x_n \mid x_1, \ldots, x_{n-1}), \tag{12}$$

and this decomposition is unique up to a relabelling of the variables. For an absolutely continuous F with strictly increasing, continuous marginal densities $F_1, \ldots, F_n$, the joint density function f is obtained from (1) as

$$f(x_1, \ldots, x_n) = c(F_1(x_1), \ldots, F_n(x_n)) \cdot \prod_{i=1}^{n} f_i(x_i). \tag{13}$$

In the bivariate case, (13) simplifies to

$$f(x_1, x_2) = c_{12}(F_1(x_1), F_2(x_2)) \cdot f_1(x_1) \cdot f_2(x_2), \tag{14}$$

where c_{12} is the pair-copula density for $F_1(x_1)$ and $F_2(x_2)$ (called *transformed variables*). For a conditional density $f(x_2 \mid x_1)$ [the second factor of (12)] it follows that

$$f(x_2 \mid x_1) = c_{12}(F_1(x_1), F_2(x_2)) \cdot f_2(x_2), \tag{15}$$

for the same pair-copula c_{12}. In the three-variate case, the third factor of (12) can be written as

$$f(x_3 \mid x_1, x_2) = c_{31|2}\left(F_{2|1}(x_2 \mid x_1), \times F_{1|2}(x_1 \mid x_2)\right) \cdot f(x_3 \mid x_2), \tag{16}$$

where $c_{31|2}$ is the pair-copula density for the transformed variables $F_{2|1}(x_2 \mid x_1)$ and $F_{1|2}(x_1 \mid x_2)$. An alternative decomposition of the third factor in (12) is

$$f(x_3 \mid x_1, x_2) = c_{32|1}\left(F_{3|1}(x_3 \mid x_1), F_{2|1}(x_2 \mid x_1)\right) \cdot f(x_3 \mid x_1). \tag{17}$$

where $c_{32|1}$ is different from $c_{31|2}$ in (16). Notice that, given a specific factorization, there are different decompositions. Decomposing $f(x_3 \mid x_1)$ in (17) further, leads to

$$f(x_3 \mid x_1, x_2) = c_{32|1}\left(F_{3|1}(x_3 \mid x_1), F_{2|1}(x_2 \mid x_1)\right) \times c_{13}(F_1(x_1), F_3(x_3)) \cdot f_3(x_3). \tag{18}$$

Then we may generalize the factorization of each term in (12) into pair-copulas and a marginal conditional distribution,

$$f(x_i \mid \mathbf{v}) = c_{ij|\mathbf{v}_{-j}}\left(F\left(x_i \mid \mathbf{v}_{-j}\right), F\left(v_j \mid \mathbf{v}_{-j}\right)\right) \cdot f\left(x_i \mid \mathbf{v}_{-j}\right), \tag{19}$$

for a d-dimensional vector $\mathbf{v}$, where v_j is one arbitrarily chosen component of $\mathbf{v}$ and $\mathbf{v}_{-j}$ denotes the vector $\mathbf{v}$ excluding the component v_j. Combining (12) and (19), any multivariate density function $f(x_1, \ldots, x_n)$ can be expressed as a product of pair-copulas (applied on unconditional and conditional distribution functions) and marginal densities. Such a decomposition is called *pair-copula construction.*

For PCC, marginal conditional distributions of the form $F(x_i \mid \mathbf{v})$ are needed. For every v_j in $\mathbf{v}$, $F(x_i \mid \mathbf{v})$ can be written as

$$F(x_i \mid \mathbf{v}) = \frac{\partial C_{ij|\mathbf{v}_{-j}}\left(F\left(x_i \mid \mathbf{v}_{-j}\right), F\left(v_j \mid \mathbf{v}_{-j}\right)\right)}{\partial F\left(v_j \mid \mathbf{v}_{-j}\right)}, \tag{20}$$

where $C_{ij|\mathbf{v}_{-j}}$ is a bivariate copula distribution. For parametric pair-copula densities and univariate conditioning set v, when x and v are uniform [note that, in this case, $f(x) = 1$, $f(v) = 1$, $F(x) = x$ and $F(v) = v$], (20) simplifies to

$$F(x \mid v) = \frac{\partial C_{xv}(F(x), F(v))}{\partial v}. \tag{21}$$

As we noted before, given a specific factorization, there are different decompositions. Actually, the number of copula constructions grows exponentially with n (in fact, it is $n!/2$). In order to reduce the number of such decompositions, in [2,3] a graphical representation is introduced using densities and a sequence of nested trees. The class of multivariate copulas constructed in such a way is called regular vines. The so-called Dvines is a subclass of regular vines. In this work, we focus on this copula.

Definition 2 (*Rvine, Dvine*) A vine is a sequence of $n-1$ nested trees where:

(i) The tree T_1 is a tree on nodes 1 to n.
(ii) Tree T_j, $j = 1, \ldots, n-1$, has $n+1-j$ nodes and $n-j$ edges.

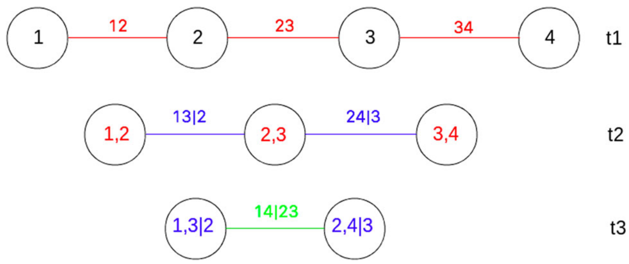

Fig. 3 The hierarchical tree structure of the four-dimensional Dvine. Edges in the tree correspond to pair-copula densities denoted by the edge labels. The Dvine consists of three trees, nine nodes and six pair-copulas

(iii) Edges in tree T_j become the nodes of tree T_{j+1}.

An Rvine is a vine that satisfies that two nodes in tree T_{j+1} can be joined by an edge only if they share a common node in T_j. A Dvine is an Rvine for which each tree T_j is connected to at least another node but no more than two (path structure).

The Dvine representation for the multivariate density $f(x_1, \ldots, x_n)$ can be expressed as

$$\prod_{j=1}^{n-1}\prod_{i=1}^{n-j} c_{i,i+j|i+1,\ldots,i+j-1}\left(F\left(x_i|x_{i+1},\ldots,x_{i+j-1}\right),\right.$$
$$\left.F\left(x_{i+j}|x_{i+1},\ldots,x_{i+j-1}\right)\right)\cdot\prod_{k=1}^{n} f(x_k), \tag{22}$$

where j and i identify the trees and the edges in each tree, respectively. In the first tree, there are $n-1$ unconditional pair copulas whose arguments are two unconditional distribution functions. In the remaining trees, the arguments of the conditional pair-copulas are conditional distribution functions.

Figure 3 shows the tree representation of a four-dimensional Dvine. It factorizes the multivariate copula c_{1234} $(F_1(x_1), F_2(x_2), F_3(x_3), F_4(x_4))$ as

$$\begin{aligned} c_{1234} = {} & c_{12}(F(x_1), F(x_2)) \cdot c_{23}(F(x_2), F(x_3)) \\ & \times c_{34}(F(x_3), F(x_4)) \cdot c_{13|2}(F(x_1|x_2), \\ & F(x_3|x_2)) . c_{24|3}(F(x_2|x_3), F(x_4|x_3)) \\ & \times c_{14|23}(F(x_1|x_2, x_3), F(x_4|x_2, x_3)) . \end{aligned} \tag{23}$$

In the next section we introduce the vine copula-based classification approach proposed in this work, and describe the procedures to estimate such a model.

4 Vine copula classifiers

In this section, we introduce the vine copula classifier in the framework of supervised probabilistic classifiers. We consider that our dataset has been generated from an unknown probabilistic distribution $p(\mathbf{x}, \mathbf{w})$ where w is a discrete variable that represents the class labels, $w = (1, \ldots, k)$ where k represents the number of class labels; $p(w \mid \mathbf{x})$ is the probability of the instance $\mathbf{x} = (x_1, \ldots, x_n)$ being in the class w; and $p(w)$ is the probability of the class w.

In our proposal, we assume that all the classes have the same probability and the conditional probability $p(\mathbf{x} \mid w)$ is defined by a vine copula as

$$\prod_{j=1}^{n-1}\prod_{i=1}^{n-j} c_{i,i+j|i+1,\ldots,i+j-1}\left(F\left(x_i|x_{i+1},\ldots,x_{i+j-1}\right),\right.$$
$$\left.F\left(x_{i+j}|x_{i+1},\ldots,x_{i+j-1}\right) \mid w\right)\prod_{k=1}^{n} f(x_k \mid w), \tag{24}$$

where the dependence between the variables and the class is expressed by the Dvine copula model (22).

For each class w, a vine copula is used to estimate the probability distribution of the training samples that belong to the class. Then, the learned models are used to predict the probability of the unlabeled instances; and for each instance, the class whose corresponding model has assigned the highest probability to the instance is selected. In other words, (24) is used to determine the most likely class label w^*, given a vector of features $\mathbf{x}$, such that $w^* = \max_w p(w \mid \mathbf{x} = x_1, \ldots, x_n)$.

4.1 Adding flexibility to vine copula classifiers

As in other probabilistic classifiers, a Dvine classifier should learn one $p(\mathbf{x} \mid w)$ per class from the corresponding training instances. The dependence between the variables and the class represented by each $p(\mathbf{x} \mid w)$ as vine copulas can have different complexities (according to the number of trees and the types of pair-copulas). In our simplest proposal, the Dvine copulas learned for each class have the same number of trees and type of pair-copulas. This variant is called homogeneous Dvine classifiers. A more flexible approach in which, for each class, Dvines with different numbers of trees and copulas are learned is also introduced. It results in seven types of homogeneous and one heterogeneous classifier. That is to say, in this work we define eight variants (types) of classifiers that belong to one of the following groups:
Homogeneous Dvine classifiers: In our simplest proposal, one Dvine-based model is learned for each class and all the vines have the same number of trees and all pair-copulas are of just one family. Four variants of classifiers belong to this group:

- Dvine-P is the simplest classifier considered in this work. In this case, only product (P) copulas are fitted, so that $p(\mathbf{x} \mid w) = \prod_{i=1}^{n} f_i(x_i \mid w)$.

– Dvine-t1-N, Dvine-t2-N, Dvine-t3-N learn one Dvine per class and all the vines have either one (t1), two (t2) or three trees (t3), respectively; just normal (Gaussian) pair-copulas are fitted.
– Dvine-t1-Sel, Dvine-t2-Sel, Dvine-t3-Sel are more flexible types of homogeneous classifiers than those listed above because pair-copulas are selected (Sel) individually for each term from a catalogue of copula families. We call such a Dvine a mixed Dvine. The number of trees is also the same for all Dvines.

Heterogeneous Dvine classifiers: For each class $p(\mathbf{x} \mid w)$ may have different numbers of trees and pair copulas may belong either to the same or different copula families. Denoted as het-Dvine, this heterogeneous Dvine classifier is the most flexible variant designed in this work.

4.2 Dvine copula estimation

To facilitate statistical parameter estimation of higher dimensional Dvines densities (22), we apply the two-step inference function for margins (IFM) method [17,18]. According to the IFM method, the parameters of the marginal distributions are estimated separately from the parameters of the vine copula. In other words, the estimation process is divided into two steps:

First, the parameters of the marginal distributions are estimated directly from the *original data* (our training set). The data are transformed into uniform variables in (0, 1) (*transformed data*) by evaluating the previously estimated marginal distributions $F_1, \ldots, F_n$ (i.e., recall that $U_i = F_i(X_i)$, $i = 1, \ldots, n$ is a uniform random variable). Usually, parametric distributions are used to model the marginals. All marginals in (24) are modeled with Gaussian distributions, so we compute the sample means and standard deviations from the training data.

Second, the transformed (uniform) data obtained in the first step is used to estimate the Dvine copula. The estimation procedure proceeds sequentially starting by defining the first tree, continuing with the second tree, and so on. It can be either fully or partially specified according to certain criteria to truncate the structure [1,7] (we explain this idea later in this section). Aiming to explicitly model the most important bivariate relationships, it is recommended to represent the strongest dependencies in the first tree [1]. As a measure of the strength of bivariate relationships, we use the Kendall's tau. This concordance measure is especially useful when combining different copula families.

The following procedure summarizes the main steps to estimate a Dvine copula model:

1. Compute the Kendall's τ for all pairs of variables using the original data.
2. Choose an order of the variables in order to select the structure of the first tree.
3. Determine which pair-copulas to use in the first tree from the training data. Apply some test or measure to select the bivariate copulas that best fits the bivariate dependence.
4. Compute conditional distribution functions (20) using the bivariate copulas fitted in the previous tree in order to obtain the data needed for step 5.
5. Determine the pair-copulas to use in the next tree using data obtained in step 4 in the same way as step 3.
6. Repeat steps 4 and 5 until the Dvine is (fully or partially) specified.

In the following paragraph, we discuss some of the steps needed for the estimation of the Dvine copula model in detail.

Step 1 The inversion of Kendall's tau method is one of the most common methods used to estimate the parameters of bivariate copulas from data. This method works with only one-parameter bivariate copulas [30]. Kendall's τ is based on ranks and its empirical version is given by

$$\tau = \frac{\text{Concordants} - \text{Discordants}}{n(n-1)/2} \tag{25}$$

where Concordants and Discordants denote the number of concordant and discordant pairs, respectively. We say that two pairs (X_i, Y_i), (X_j, Y_j) are concordant if $(X_i - X_j)(Y_i - Y_j) > 0$, and discordant when $(X_i - X_j)(Y_i - Y_j) < 0$. As noted in [11], the case $(X_i - X_j)(Y_i - Y_j) = 0$ occurs with probability zero under the assumption that X and Y are continuous [11].

Table 1 shows analytical expressions of Gaussian, Clayton, and Gumbel copula's parameter as a function of Kendall's τ.

Step 2 For Dvines, the structure of the first tree completely determines the structure of the remaining $n-1$ trees. Therefore, the task of selecting an appropriate order of the variables is accomplished just once. Having said that, the first tree is selected in such a way that the strongest pairwise dependencies are modeled explicitly. To find an appropriate ordering of the variables, we employ the "cheapest insertion heuris-

Table 1 Copula's parameter as a function of Kendall's τ

Bivariate copula	Copula's parameter subjected to Kendall's τ
Gaussian	$\rho = \sin\left(\frac{\pi}{2}\tau\right)$
Clayton	$\delta = 2\tau/(1-\tau)$
Gumbel	$\delta = 1/(1-\tau)$

tic" that tries to find a path for which the sum of Kendall's τ associated to the edges is maximized [31].

Steps 3 and 5 The main advantage of vine copulas is their ability to represent diverse dependence structures. An appropriate Dvine-based classifier should take advantage of this property by selecting the pair-copula types that properly express the bivariate correlation, instead of fitting a bivariate copula of a unique, a priori defined, family of copulas for all linked pairs. According to this, in Step 1, we use the Cramer-von Mises statistics, $S_m = \sum_{j=1}^{m}\left(C_{\mathrm{e}}\left(u_j, v_j\right) - C_\theta\left(u_j, v_j\right)\right)^2$. In this expression, C_{e} and C_θ represent the empirical and parametric copulas, θ is the parameter of C_θ, and m is the sample size. Therefore, from a predefined group of bivariate copulas (candidate copulas), the copula C_θ that minimizes S_m is chosen [12].

Step 6 A Dvine copula utilizes $n(n-1)/2$ pair-copulas to express an n-dimensional multivariate density in (22). Check in high-dimensional practical applications, it is convenient to truncate the Dvine structure (e.g., as proposed in [1,4]). As usual, in hierarchical modeling, a model can be simplified if the initial factorization of the joint density takes advantage of the assumed conditional independence: When a vine is truncated at a certain tree, all the pair-copulas in the subsequent trees are assumed to be product copulas. Let us consider the four-dimensional Dvine decomposition in (23). If we truncate the structure at the second tree, we have that

$$c_{13|2}\left(F\left(x_1|x_2\right), F\left(x_3|x_2\right)\right) = 1$$
$$c_{24|3}\left(F\left(x_2|x_3\right), F\left(x_4|x_3\right)\right) = 1$$
$$c_{14|23}\left(F\left(x_1|x_2, x_3\right), F\left(x_4|x_2, x_3\right)\right) = 1;$$

therefore, the pair copula decomposition in (23) simplifies to

$$\begin{aligned} c_{1234} &= c_{12}\left(F\left(x_1\right), F\left(x_2\right)\right) \cdot c_{23}\left(F\left(x_2\right), F\left(x_3\right)\right) \\ &\quad \times c_{34}\left(F\left(x_3\right), F\left(x_4\right)\right) \cdot 1.1.1 \end{aligned} \tag{26}$$

by taking advantage of assumed conditional independence that is expressed by the product copula [1]: for any vector of variables $\mathbf{V}$ and two variables X, Y, it holds that X and Y are conditionally independent given $\mathbf{V}$ if and only if $c_{xy|\mathbf{v}}\left(F_{x|\mathbf{v}}\left(x|\mathbf{v}\right), F_{y|\mathbf{v}}\left(y|\mathbf{v}\right)\right) = 1$.

5 Mind reading problem

The mind decoding problem involves different multidisciplinary approaches, from the conception of the experiments and a preliminary analysis of the mental mechanisms implicated, to the decision on the type of signals to be investigated and the recording devices to employ. In this paper we approach the MRP problem from a pure machine learning perspective, in which our goal is to address the classification problems involved in the MRP. Nevertheless, to understand the particular characteristics of the classification problems addressed, some background of the experimental conditions is needed. We provide this relevant information in this section. We also make a brief review of previous work on classification approaches to the mind decoding and other related problems. Some special attention is given to the classification methods that were previously applied to the data used in this paper.

5.1 Description of the original experiment

We focus on a particular publicly available MRP benchmark produced by the Mind Reading Challenge Competition [20]. It consists of inferring the type of video stimulus shown to the subject from MEG brain signals (recorded from a single subject in two experimental sections at different time-points). A successful brain decoding classifier should be able to recognize which type of video (among the five possible) the subject has watched by analyzing the brain signals. All videos were presented without audio, and five different types of stimuli were used:

Class 1: Artificial Screen savers showing animated shapes or text.
Class 2: Nature Clips from nature documentaries showing natural scenery such as mountains or oceans.
Class 3: Football Clips taken from European football matches of "La Liga" in Spain.
Class 4: Mr.Bean Clip from the episode "Mind the baby, Mr. Bean" of the Mr. Bean television series.
Class 5: Chaplin Clip from the "Modern Times" feature film.

MRP is considered a very challenging problem for the classification methods. The brain data are highly noisy by virtue of the variability of brain signals recorded in different sessions (in particular, two different days) which immediately implies the variability between the distribution of the training and test data. The main consequence of this scenario is that the classifiers learned using the training data may have a very low prediction accuracy on the test data. Another source of difficulty is the dimension of the problem: 408 features extracted from the recorded brain signals.

In this experimental benchmark, the training and test data are taken from two different sessions: 677 training examples were recorded the first day and 653 examples the second day; 50 samples for the second day will be used for the training to facilitate the modeling of possible variations in the data distribution between the 2 days.

5.2 Previous work

Much effort has been devoted to solve the mind decoding problem (MDP) in the context of research on brain–computer interfaces (BCIs) [21,38]. BCIs are devices conceived to translate electrical signals into commands without the need for motor intervention. Different voluntary and involuntary mental processes can serve as a basis for implementing BCIs, but in most of the cases a decoding component is required to decode the stimulus received by the subject or his intended command. As part of this decoding component, classification algorithms are implemented. A variety of classification algorithms have been used to analyze brain data in the context of BCI applications [22]. They include linear discriminant classifiers (LDA) [10], support vector machines (SVMs) [5,26], neural networks (NNs) [13], and other classification methods [25,27,29,35]. For a survey of classification algorithms applied to BCI, [22] can be consulted.

Until recently, an obstacle limiting the advance of research on MRPs was the lack of available brain recording databases where machine learning approaches could be tested. However, in the past few years a number of brain decoding or mind reading competitions have provided the needed benchmarks to evaluate and develop more accurate classifiers for this problem. In particular, the Mind Reading Competition [20] allowed the comparison of nine different approaches to solve this challenging classification task.

We briefly analyze the approaches presented by the top three teams, to which we compare our approach in the section of experiments. The classification accuracies achieved by these algorithms are shown in Table 6.

Huttunen et al. [14] presented an algorithm based on regularized logistic regression. The authors explored a number of features and decided to use the mean and the slope. In a later work in [15], the authors extended the analysis of the classifier and compared the same approach in other datasets. An essential aspect of the success of the contribution was the error assessment. In the proposed method, data from the first and second days were combined. Samples were weighted such that data from the second day had a higher weight in the cost function used to learn the classifier. Also, to estimate the error, large computational resources were employed. The training data were split in two parts, and this was done in hundreds of ways. For each split a dedicated processor was assigned the k-fold cross-validation task. The computation was done on a grid of computers. We emphasize two main differences between the solution presented in [14] and our approach. First, we do not give different weights to samples recorded the second day in order to learn the vines. Second, for assessing the error we conducted k-fold cross-validation.

Santana et al. [34] used an ensemble of three types of classifiers: regularized multi-logistic regression, regression trees, and an affinity propagation [9] based classifier. The authors also proposed the combination of the classifiers learned from different types of features extracted from raw data, channel correlations, mutual information between channels, and channel interaction graphs. The computation of this large number of features is a costly process. In [34] it is not discussed whether the classification accuracies obtained by the ensemble approach were due to the synergy between the different types of features used or to the combination between different classifiers. The method we propose in this paper relies on a smaller number of features than the one used in [34]. We consider that our approach is also simpler because it considers a single type of classifiers: vines copulas.

The approach proposed by Jylanki et al. [19] used a representation based on power features from the filtered MEG signals. In addition, linear one-versus-rest logistic classifiers were applied for dimensionality reduction of the MEG gradiometer channel space. Nonlinear Gaussian process (GP) multi-class classifier with a squared exponential covariance function [28] was applied to predict the class from the features. Although the GP approximation is suitable for theoretical analysis, the learning process can be computationally demanding and sensitive to numerical instabilities. However, these aspects of the solution were not discussed in [19].

Although we have not found previous publications on the use of vines as classifiers, vines were used for regression in an experimental framework in which covariate shift.

6 Experimental framework

In this section we empirically investigate the performance of the vine copula classifiers in the solution of the MRP. First, we present a description of the experimental framework, explaining in detail the classification scenarios considered and how the training and test data were selected in each case. Then we present a comparison between different variants of vine copula classifiers according to the complexity of Dvines. Finally, we compare the approach introduced in this paper with the results of other classifiers applied to MRP.

6.1 MRP classification scenarios

As previously mentioned, in this experimental benchmark, the training and test data were recorded from two different sessions. One important question is to determine whether the potential variations between the distribution of the training and test sets affects the accuracies obtained by the classifiers. Therefore, we considered two different scenarios for the validation of the classifiers:

Training 1: The training data contain examples recorded only the first day.

Table 2 Comparison between Dvine-based classifiers in the two different scenarios: Training 1 and Training 2

Classifier	Global accuracy			
	Training 1		Training 2	
	5-fold-validation	Classification	5-fold-validation	Classification
Dvine-P	62.00	49.15	62.71	**52.22**
Dvine-t1-N	72.36	46.09	71.81	**48.85**
Dvine-t2-N	81.40	52.52	79.98	**59.57**
Dvine-t3-N	82.07	52.67	79.79	**62.02**
Dvine-t1-Sel	61.06	48.08	68.16	**52.67**
Dvine-t2-Sel	76.72	53.44	73.47	**60.79**
Dvine-t3-Sel	77.40	54.21	75.03	**62.63**

Accuracies are computed in the validation and classification steps
Highest accuracy achieved in each class are highlighted in bold

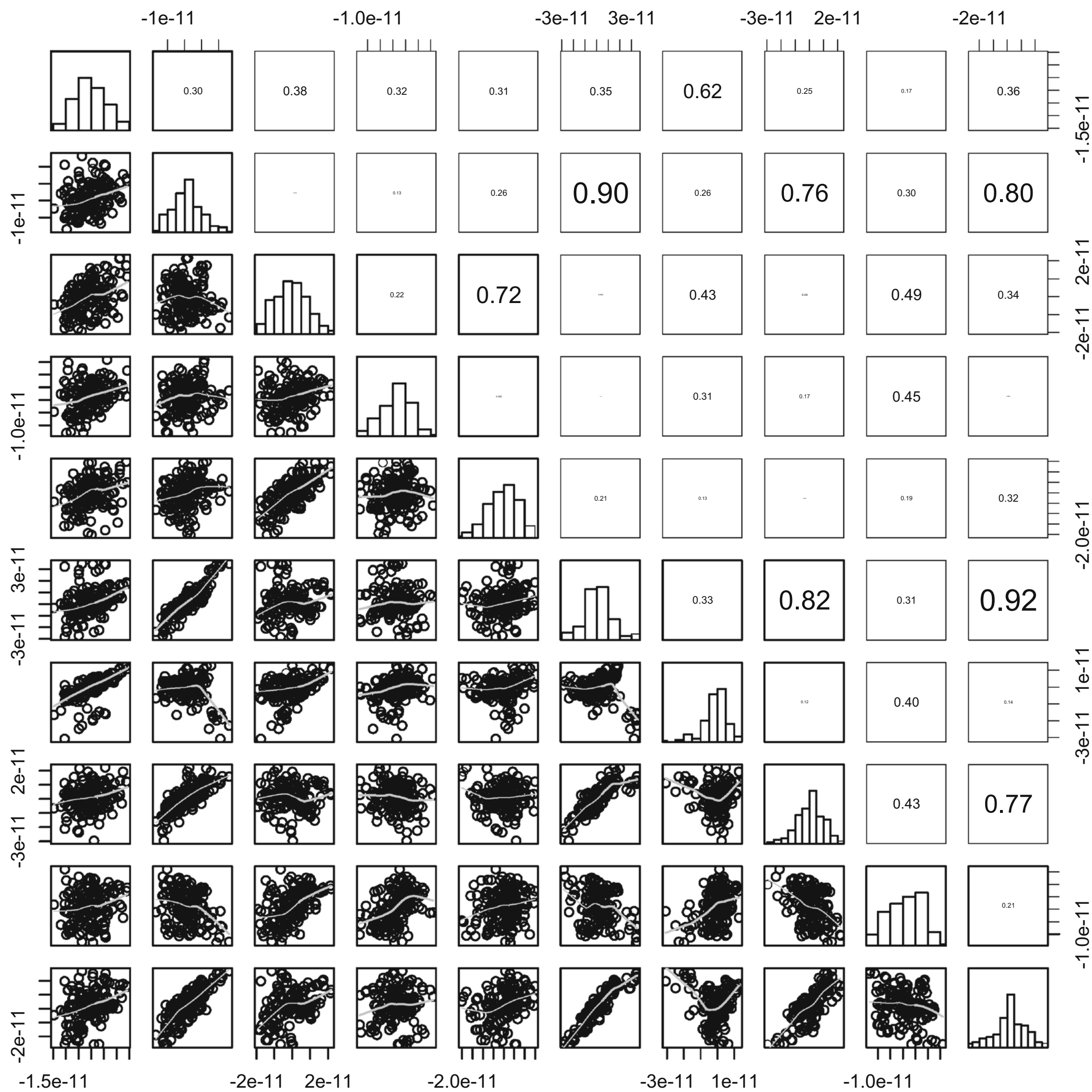

Fig. 4 Pairwise scatter plot of 10 arbitrary features from a total of 408 for the Class 1 (Artificial)

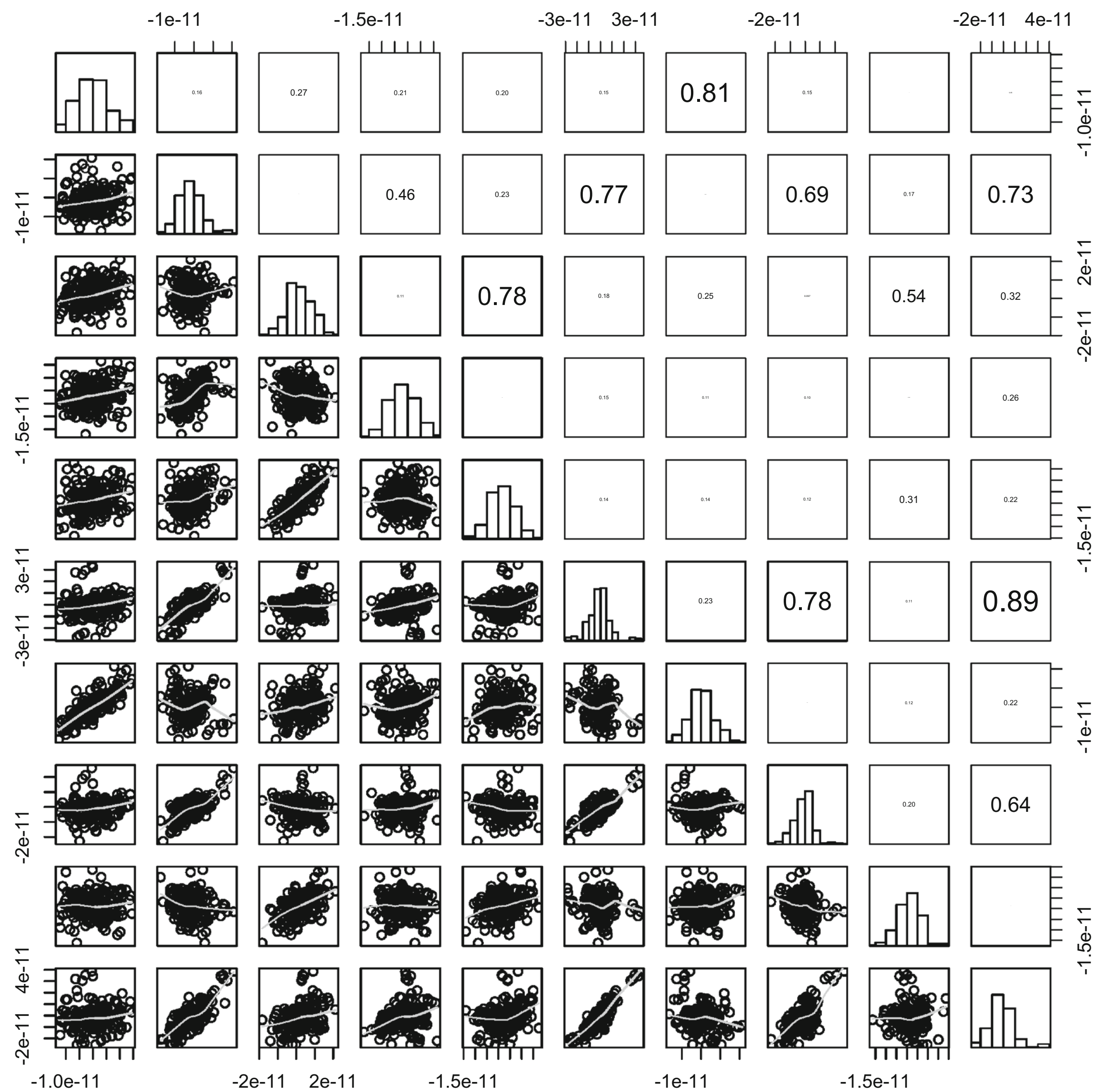

Fig. 5 Pairwise scatter plot of 10 arbitrary features from a total of 408 for the Class 2 (Nature)

Training 2: The training data are mixed, containing the 677 examples recorded the first day plus 50 examples of the second day

6.2 Comparison of vine copula classifiers

In the classification approach introduced in this work, the conditional probability $p(\mathbf{x} \mid w)$ is defined by (24), which means that the dependence between the features and the class is expressed by a Dvine copula. Under this classification scheme, as we have five stimulus categories, we learn one Dvine copula per class from the corresponding training data using the estimation procedures described in Sect. 4.2. Therefore, a single Dvine classifier consists of five Dvine copulas of 408 variables.

Furthermore, taking into account possible differences between the distribution training and testing data, we have decided to carry out a fivefold cross validation with the training data in order to highlight these differences. In this context, we will evaluate the performance of the classifiers at two different phases (steps):

– At validation step, a 5-fold cross validation is carried out using only the training data.
– At classification step, a Dvine copula classifier is learned from the training data; then, the instances from the test data are classified using the Dvine classifier.

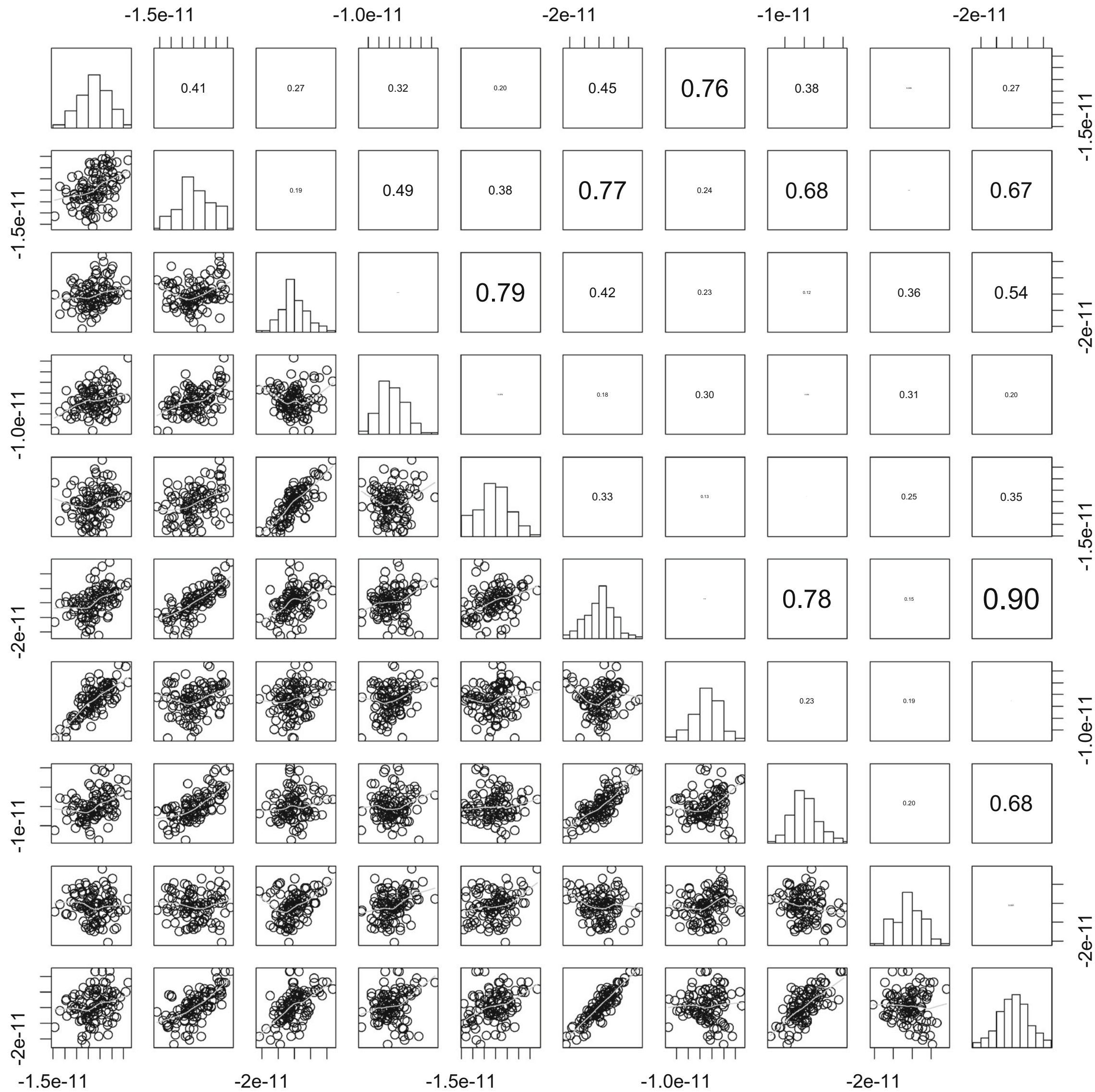

Fig. 6 Pairwise scatter plot of 10 arbitrary features from a total of 408 for the Class 3 (Football)

In the experiments, we compare the classification accuracies with the eight Dvine classifier variants described in Sect. 4.1. For both variants of classifiers, homogeneous mixed Dvine (Dvine-t1-Sel, Dvine-t2-Sel, Dvine-t3-Sel) and heterogeneous het-Dvine, the catalogue of candidate copulas consists of Product, Normal, Clayton, and Gumbel families.

The main criteria that we use to evaluate the performance of the classifiers are the global and class accuracies computed from the confusion matrix.

7 Results and discussion

This section presents and discusses the experimental results.

7.1 Comparison between Training 1 and Training 2 datasets

In this experiment we compare the performance of the seven types of Dvine-based classifiers when using Training 1 and Training 2. It is expected that the performance of the classifiers will be more accurate for Training 2. The accuracy results shown in Table 2 (columns 3 and 5) confirm our hypothesis. For all Dvine-based classifiers, accuracies are higher for Training 2. One possible reason for this behavior is that, by including data from the second day, the classifiers learn those dependence patterns that remain unchanged across sessions. Also notice that the number of available samples is higher in Training 2. Therefore, additional information

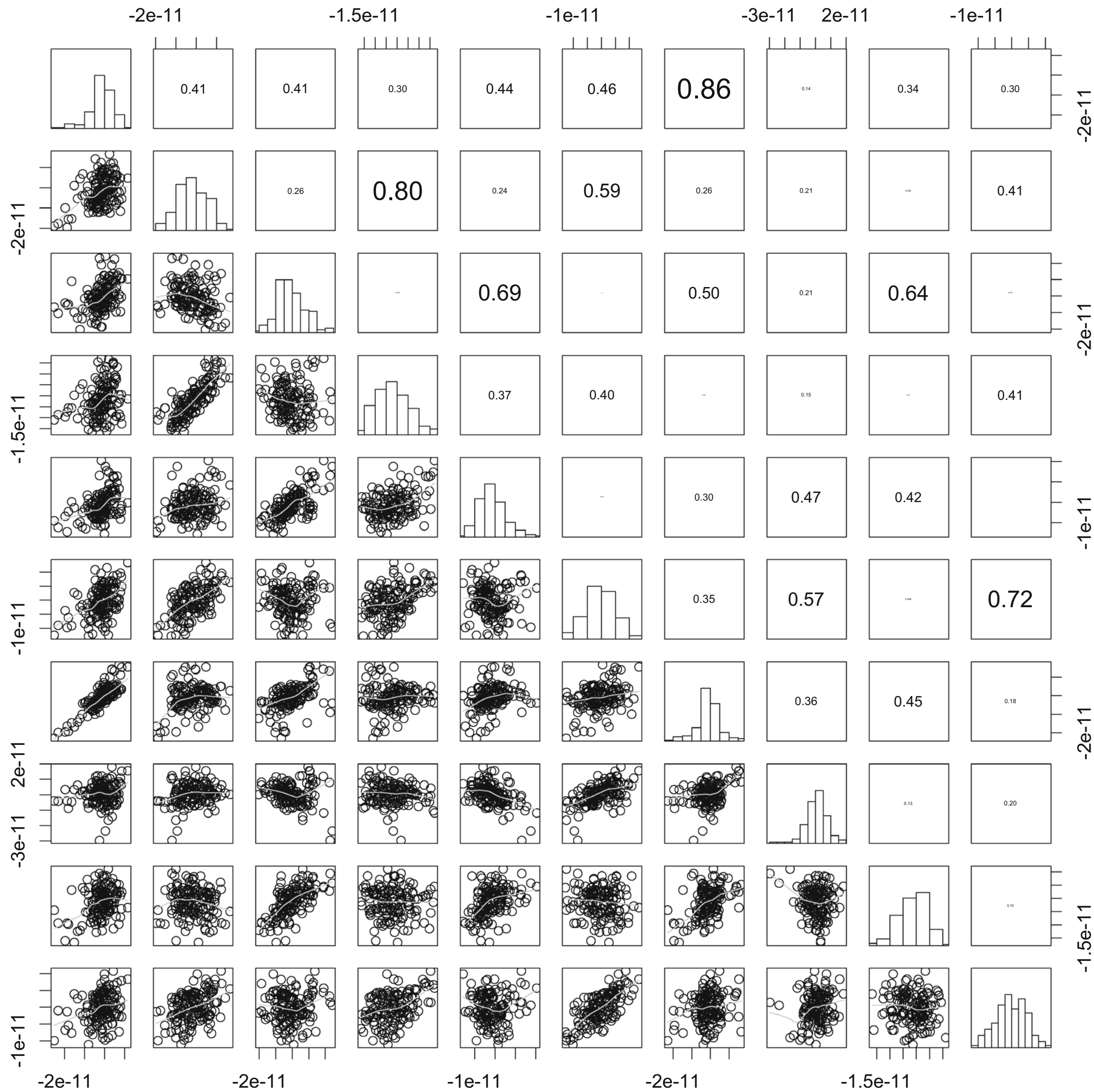

Fig. 7 Pairwise scatter plot of 10 arbitrary features from a total of 408 for the Class 4 (Bean)

may contribute to improve the classification accuracy. From here on, the next experiments are performed just for Training 2.

7.2 Dvine copulas are able to capture the stimulus class dependencies

In this section we test whether Dvine-based classifiers can recognize the dependence patterns that arise in our problem.

A simple visual inspection of pairwise scatter plots (Figs. 4, 5, 6, 7, 8) for 10 arbitrary features (from a total of 408) reveals a great variety of bivariate correlation patterns in all classes: there exists positive/negative and weak/strong correlated pairs (small and large absolute values of Kendall's tau located in the upper-triangle of the pair-matrix plots).

The global and per class accuracy at validation and classification steps for Training 2 are displayed in Table 3. The main results are listed as follows:

– Essentially, all classifiers performed well. At validation-step the observed performance of all classifiers was better than at classification-step.
– At classification-step, Dvine-t3-Sel achieves the highest global accuracy among all classifiers. It is also the most accurate in three classes (c2, c4, c5).
– The accuracy increases with the number of trees.
– The most interesting result is that, at validation-step, the classifiers based only on normal pair-copulas attain better results (globally and per-class) than those that use pair-copula selection. However, at classification-step exactly

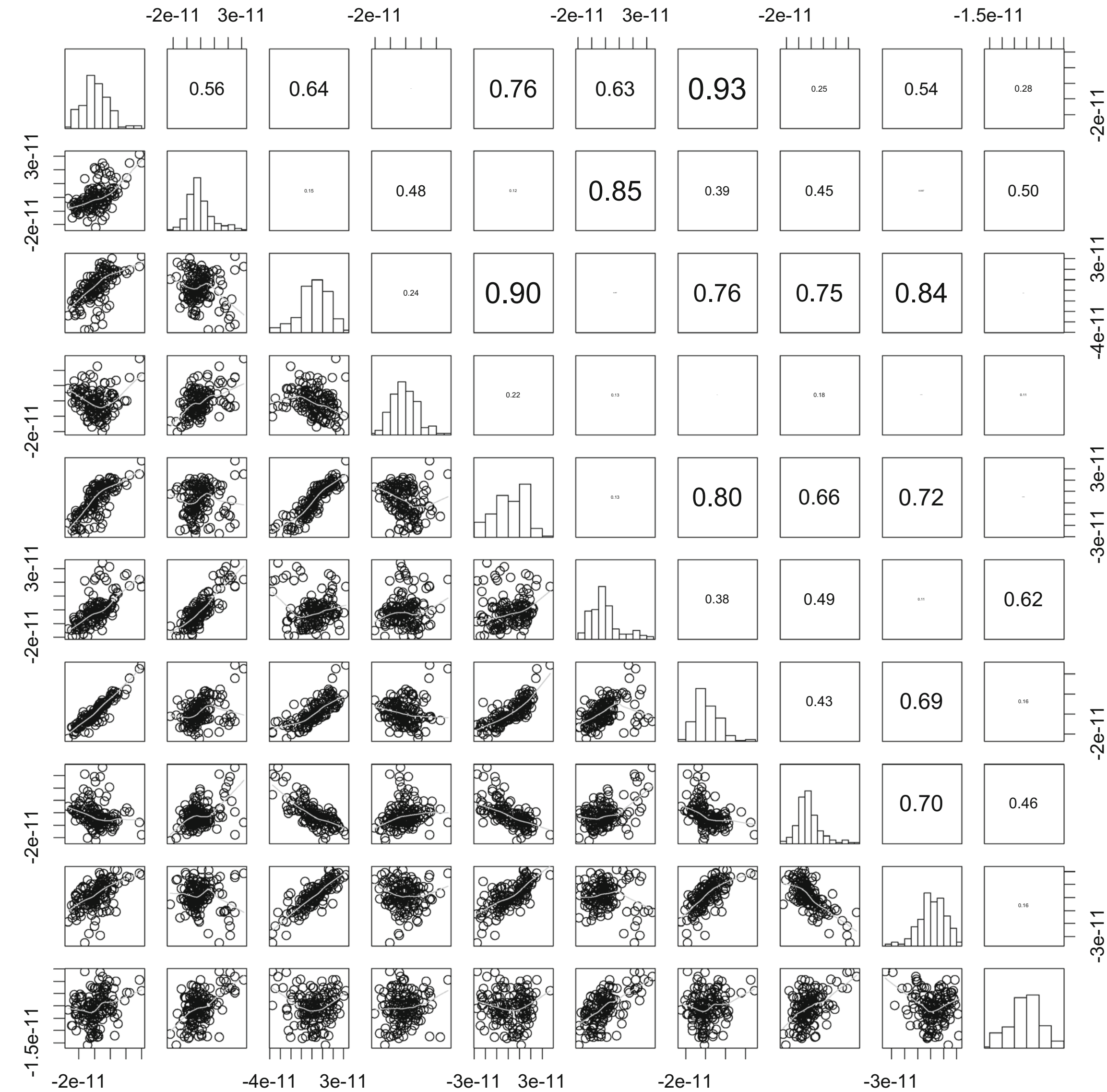

Fig. 8 Pairwise scatter plot of 10 arbitrary features from a total of 408 for the Class 5 (Chaplin)

the opposite is valid. Notice, for example, that Dvine-t2-N reaches a value of global accuracy of 79.98 at validation step, whereas at classification-step it decreases in 20.41 units. Furthermore, the two most "affected" classes, with respect to the same classifier, were c2 and c4. Moreover, the classifiers based on mixed copulas perform better than those based on normal ones.

The covariate shift has a more negative effect on the normal classifiers than on the mixed ones as a result of a higher capacity to capture the variations in the data, given by the covariate shift phenomena. Hence, we would prefer heterogeneous classifiers over the homogenous ones, despite the fact that they are computationally more expensive.

7.3 Heterogeneous Dvine classifier to MRP

So far, we have modeled the dependence structure of the five classes with the same Dvine model for a tested classifier. However, this "homogeneous" approach does not necessarily have to be the most appropriate. Notice, for instance, that the global accuracy reached by Dvine-t3-Sel (62.63 %), Table 3, is higher than the value obtained by Dvine-t1-Sel (52.67 %). However, the second classifier obtains better accuracy than the former in two of the classes: in c1: 55.33 vs. 48.66 %, while in c3: 51.96 vs. 35.29 % (Table 4). This behavior suggests a more flexible approach that combines different Dvine-based models (with regard to the number of trees as well as the pair copula families, which is exactly the

Table 3 Global and class accuracy at validation and classification steps for Training 2

Classifiers	c1	c2	c3	c4	c5	Global
Validation step						
Dvine-P	79.81	53.71	57.10	35.01	87.94	62.71
Dvine-t1-N	58.33	67.08	62.61	86.72	84.31	71.81
Dvine-t2-N	64.50	79.72	**72.61**	**94.31**	88.79	**79.98**
Dvine-t3-N	**67.50**	**83.19**	66.90	92.41	**88.96**	79.79
Dvine-t1-Sel	60.00	61.80	50.23	85.68	83.10	68.16
Dvine-t2-Sel	59.30	78.19	50.71	92.75	86.37	73.47
Dvine-t3-Sel	59.33	81.52	52.61	96.20	85.51	75.03
Classification step						
Dvine-P	**76.70**	35.76	47.05	16.80	82.40	52.22
Dvine-t1-N	62.66	47.01	49.01	14.40	68.80	48.85
Dvine-t2-N	54.66	**64.90**	47.05	44.00	84.80	59.57
Dvine-t3-N	50.00	64.23	50.98	59.20	85.60	62.02
Dvine-t1-Sel	55.33	47.68	**51.96**	32.00	76.80	52.67
Dvine-t2-Sel	51.33	60.92	36.27	68.00	84.80	60.79
Dvine-t3-Sel	48.66	**64.90**	35.29	**76.80**	**86.40**	**62.63**

Highest accuracy achieved in each class are highlighted in bold

Table 4 Confusion matrices at classification step for Training 2

Class	Dvine-t1-N					Dvine-t2-N					Dvine-t3-N				
	c1	c2	c3	c4	c5	c1	c2	c3	c4	c5	c1	c2	c3	c4	c5
c1	94	18	29	4	5	82	34	23	6	5	75	38	24	8	5
c2	53	71	17	5	5	33	98	8	7	5	27	97	10	12	5
c3	31	17	50	3	1	32	18	48	3	1	31	13	52	5	1
c4	22	17	41	18	27	11	12	17	55	30	7	4	9	74	31
c5	13	0	22	4	86	4	2	7	6	106	2	2	6	8	107
Class	Dvine-t1-Sel					Dvine-t2-Sel					Dvine-t3-Sel				
	c1	c2	c3	c4	c5	c1	c2	c3	c4	c5	c1	c2	c3	c4	c5
c1	83	19	33	8	7	77	34	11	19	9	71	40	11	21	7
c2	44	72	22	6	7	26	92	6	21	6	20	98	2	25	6
c3	34	9	53	5	1	32	20	37	12	1	32	19	36	12	3
c4	26	5	23	40	31	5	3	0	85	32	4	1	0	96	24
c5	9	1	11	8	96	1	1	2	15	106	1	1	2	13	108
Class	Dvine-P														
	c1	c2	c3	c4	c5										
c1	115	8	24	2	1										
c2	72	54	18	5	2										
c3	46	4	48	2	2										
c4	35	7	45	21	17										
c5	7	0	13	2	103										

rationale of the heterogeneous classifiers introduced in Sect. 4.1).

We tested all possible combinations of the seven Dvine copula types for each of the five classes described in the previous sections, that is combinations. The results summarized in Table 5 show the Dvine model combination that reaches the highest accuracy, denoted as het-Dvine classifier. The dependence structure for the classes c1 and c2 has been modeled by mixed Dvine copulas that use three trees, whereas for the classes c1, c3, c5 just normal pair-copulas have been fitted,

Table 5 Global accuracy and confusion matrix obtained by het-Dvine with Training 2

Class	het-Dvine model combination	Global accuracy
c1	Dvine-t2-N	61.33
c2	Dvine-t3-Sel	52.98
c3	Dvine-t3-N	54.90
c4	Dvine-t3-Sel	64.00
c5	Dvine-t2-N	88.00
Global accuracy of het-Dvine		**64.01**

Confusion matrix for het-Dvine

Class	c1	c2	c3	c4	c5
c1	92	16	24	12	6
c2	34	80	16	16	5
c3	34	4	56	7	1
c4	12	0	12	80	21
c5	5	1	7	2	110

Highest accuracy achieved in each class are highlighted in bold

Table 6 Global accuracy of the best four teams at Mind Reading Challenge and het-Dvine with Training 2

Team	Global accuracy
Huttunen et al.	**68.0**
het-Dvine	64.0
Santana et al.	63.2
Jylanki et al.	62.8
Tu et al.	62.2

Highest accuracy achieved in each class are highlighted in bold

with two trees for c1 and c5, and three for c3. The dependence structure of brain signals seems to be quite complex across classes. This model class combination leads to improving the accuracy of 62.63 % (Table 3, Dvine-t3-Sel) to 64.01 % (het-Dvine).

7.4 Comparison with previous approaches

In Table 6 we compare het-Dvine with the four best teams at MEG Mind Reading Competition according to the global accuracy. The results of het-Dvine rank second in the list of previous approaches. Such an accuracy has been obtained without any further processing of the data or applying more elaborated strategies for Dvine model selection (Sect. 5.2).

8 Conclusions

In this paper we have presented a general vine copula-based approach for solving supervised classification tasks. Our proposal consists of training a vine copula model per class. Afterwards, the estimated models are evaluated on examples not seen during training, which are then assigned to the class for which they obtained the highest probability.

Although brain signal classes have a diverse and complex dependence structure, Dvine-based classifiers have been able to discover them. This backs the feasibility of the proposed approach in detecting the diverse dependences hidden in multivariate brain signals. Furthermore, its capacity to use this information to accurately decode them, even under the effect of covariate shift, is evident in the results of the classifiers. In fact, the most robust classifier is the one that learns different types of Dvine-copulas per class, combines different pair-copula families, and utilizes a hierarchical structure of two and three trees to represent conditional dependencies of a high order.

Although we focused on the straightforward application of the vine model to the classification problem, there are a number of lines that should be investigated to further assess the potential of vine copulas as classifiers. We discuss some of these topics for further work here:

- The general question of how to use a priori information about the problem for learning more accurate vines should be investigated. For example, we could determine whether a priori information about the classification problem (groups, types, etc) of the variables can be effectively translated into constraints that, when used in the construction of the vines (e.g., by constraining the dependencies to be searched during the vine construction), reduce the complexity of the model learning step while keeping or even improving the accuracy. Another possibility is to take into account the characteristics of the copula models to be used (e.g. Gaussian) at the time of performing feature selection, i.e., we do not only consider the discriminatory power of the features, but also how well is the distribution of the candidate feature modeled by the sets of copula types used to learn the vines.
- One open question is whether a posteriori analysis of the vines learned for each class could be used to estimate feature-importance.
- We have mainly focused on the effect of the dependence structure of brain signals in the predictive ability of Dvine-based classifiers. We have modeled the marginal distributions of brain signals with the Gaussian distribution. However, it seems to be that they do not always follow this distribution (see the main diagonal of Figs. 4, 5, 6, 7, 8). We believe that a more effective distributional form for the margins may lead to increasing the predictive ability of our classifiers.
- Finally, one relevant question worthy of investigation is how to modify the methods used to learn vines to limit the impact of covariate shift in classification accuracy.

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